Isaac Adjei

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Electronic Engineering & Computer Science student at Aston University (Predicted First Class). Partially sighted since age two following suspected retinoblastoma - I bring focus and determination to every challenge, from bare-metal firmware and PCB design to full-stack web and AI/ML. Seeking internships, placements or part-time roles in engineering and technology.

Education

Aston University, Birmingham, United Kingdom *Oct 2024 – Jul 2028*
BEng (Hons), Electronic Engineering and Computer Science · ***Predicted: First Class***
Top 40 Finalist – Black Heritage Undergraduate of the Year Award 2026 (TargetJobs & Sky)

Stanmore College, London, United Kingdom *Sep 2022 – Jul 2024*
Pearson BTEC Level 3 National Extended Diploma in Engineering · ***D*DD (Distinction*, Distinction, Distinction)***
Best and Most Hardworking Student Award

Experience

Student Representative ***Aston Students' Union, Birmingham, UK***
Student advocacy, academic liaison *Oct 2024 – Present*

- Represent student views to academic staff and university management, facilitating communication to **improve the course experience** for the department

Smart Mobility & Digital Tech (Virtual Work Experience) ***Yunex Traffic via Springpod, Remote***
Intelligent transport systems, sensor integration, data-driven urban mobility *Aug 2025*

- Explored how **ITS and digital technology** improve air quality and urban efficiency through traffic lights design, sensor integration and data-driven transport management

Consular & Administrative Intern ***Ghana High Commission, London, UK***
Consular operations, document management, facilities coordination *Mar 2024 – Aug 2024*

- Consular: supported **passport and visa processing**, document verification and public-facing client enquiries
- Administrative & Estates: managed **facilities operations**, document handling and administrative coordination

Junior Apprentice HVAC Technician ***Massive Refrigeration Services, Accra, Ghana***
Air conditioning systems, diagnostics, field installation *Jul 2019 – Jul 2021*

- Serviced, installed and maintained **50+ air conditioning units** on-site using hand tools, drills and diagnostic meters

Student Judge – National Emerging Talent Awards 2026 ***targetjobs UK (Group GTI)***
Volunteering · employer evaluation, qualitative assessment *Feb 2026*

- Evaluated employer submissions for Best Placement or Internship Programme; **commended** for a thorough and high-quality approach

Fundraising Volunteer – 10 Days of 5K Challenge ***Cancer Research UK***
Volunteering · fundraising, personal challenge *Feb – Mar 2026*

- Completed **10 x 5km runs (50km+)** contributing to a campaign that raised over **£797,000** for cancer research

Projects

Two-Stage Audio Amplifier *2025*
C, KiCad, Proteus, LTSpice, Oscilloscope Portfolio ↗

- Designed and manufactured a **two-stage class A/B PCB from scratch**: schematic capture, layout, build and bench testing; measured **frequency response, voltage gain and clipping threshold** across both stages

4x4x4 NeoPixel LED Cube *2025*
Arduino, C++, WS2812B, Electronics GitHub ↗ | Portfolio ↗

- Non-blocking **state machine architecture** using millis() with adaptive brightness, four animation modes, debounced inputs and buzzer feedback

AstonCV – Full-Stack CV Database *2026*
PHP 8.2, MySQL, CSS, Apache, Cloudflare GitHub ↗ | Portfolio ↗

- Built from scratch with no frameworks: pure PHP 8.2, MySQL, CSS with **11 security measures** including bcrypt, CSRF, PDO and brute-force protection — deployed live at Aston University

git-unlocked – Open Source Git Course *2026*
Git, GitHub, GitLab, Markdown, Open Source GitHub ↗ | Portfolio ↗

- **170+ structured topic files** spanning beginner to advanced, covering Git, GitHub, GitLab, Bitbucket, Azure DevOps, Gitea, Forgejo and Codeberg

Technical Skills

Languages: C, C++, Python, TypeScript, JavaScript, Java, PHP, SQL, MATLAB
Embedded: Arduino, STM32, ESP32, Raspberry Pi, AVR, KiCad, Proteus, FreeRTOS
Web: Next.js, React, Node.js, Tailwind CSS, PHP 8.2, MySQL, Apache
AI / ML: TensorFlow, PyTorch, NumPy, Pandas, Scikit-Learn, OpenCV
Cloud & DevOps: AWS, Azure, Docker, Vercel, Cloudflare, GitHub Actions
Hardware: PCB Layout, Circuit Design, Soldering, Oscilloscope, I2C / UART / SPI